

The CTF challenge this writeup is for is the Faulty image 4

Firstly, we are given an image with a flag written on it which is `LITCTF{fau1ty_f0uR_12ox!}`, which when we enter it in is not the correct flag, and we are also given a tip saying certain pixels have unusually small blue values like they are clipped to 4 bits and asking us which pixels are them, and also the question tells us that the encoding information is scattered around different pixels which are the unusually small blue values pixels, these pixels hold 4 bits of data or nibbles, since letters require 8 bits of data so they hid the info in the consecutive pixels in their blue channel which we need to run a script to extract the low blue value pixel's blue value and also to decrypt or piece together the flag. The code and image are in the file where this writeup can be seen. The flag is `LITCTF{pr1m3_n1bbl3s_77zq!}`, remember to change the last line into your directory!!